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4.8 Chemical Analysis

AQA GCSE Chemistry (8462) — Paper 2

Analysts have developed a range of qualitative tests to detect specific chemicals. The tests are based on reactions that produce a **gas with distinctive properties**, a **colour change** or an **insoluble solid (precipitate)** that appears in a solution. Instrumental methods provide fast, sensitive and accurate means of analysing chemicals, and are particularly useful when the amount of chemical being analysed is small. This topic includes two required practicals: **chromatography** and **identifying ions**. Sections marked **(Chemistry only)** are not in Combined Science; content marked **(HT only)** is Higher Tier.

4.8.1 Purity, Formulations and Chromatography

4.8.1.1 Pure Substances

The word "pure" means two different things depending on who is using it.

> **In chemistry**, a **pure substance** is a **single element or compound**, not mixed with any other substance.
>
> **In everyday language**, "pure" can mean a substance that has had nothing added to it, so it is unadulterated and in its natural state — e.g. "pure milk" or "pure orange juice". These are actually **mixtures** of many substances.

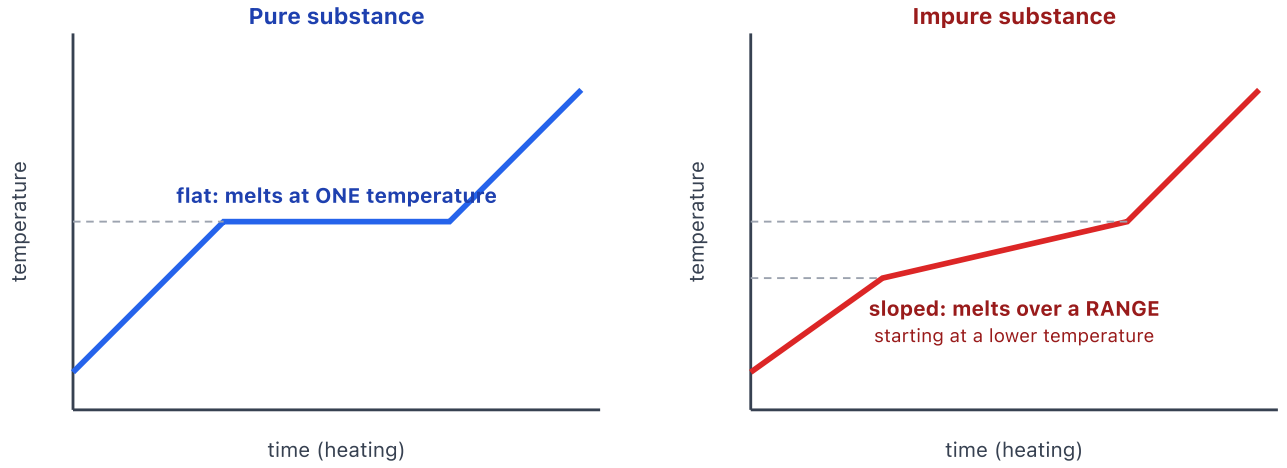
Using Melting and Boiling Points to Test Purity

Pure elements and compounds **melt and boil at specific (fixed) temperatures**. Melting point and boiling point data can be used to distinguish pure substances from mixtures.

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Feature	Pure substance	Impure substance (mixture)
Melting point	Sharp, fixed value (melts at one temperature)	**Lower** than the pure substance and melts over a **range** of temperatures
Boiling point	Sharp, fixed value	Usually **higher** than the pure substance and boils over a **range**
Example	Pure water: melts at exactly 0 °C, boils at exactly 100 °C (at standard pressure)	Sea water: freezes below 0 °C (around −2 °C) and boils above 100 °C

Melting a pure vs an impure solid



Exam Tip: If an exam question gives you a melting point and asks whether a sample is pure, compare it with the data-book value. If the sample melts **at** the expected temperature, sharply, it is pure. If it melts **below** the expected value or **over a range**, it is impure. Always quote the numbers in your answer.

4.8.1.2 Formulations

A **formulation** is a **mixture** that has been designed as a **useful product**. Many products are complex mixtures in which each chemical has a particular purpose.

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Formulations are made by mixing the components in **carefully measured quantities** to ensure that the product has the required properties. Changing the proportions changes the properties, so manufacturers follow a precise "recipe".

Formulation	Typical components and their purposes
Paints	Pigment (colour), binder (sticks pigment to surface and forms a film), solvent (makes it spreadable; evaporates), additives (e.g. to speed drying or prevent mould).
Medicines	Active ingredient (the drug), fillers/bulking agents (make a tablet a sensible size), binders, flavourings, colourings, coatings.
Fuels	Petrol is a mixture of hydrocarbons plus additives (e.g. anti-knock agents, detergents to clean the engine).
Cleaning agents	Surfactants (detergents), water softeners, enzymes, fragrances, bleaching agents.
Alloys	Metal mixtures with precise percentages, e.g. stainless steel (iron, chromium, nickel, carbon).
Fertilisers	Precise ratios of nitrogen, phosphorus and potassium compounds (NPK).
Foods	Ingredients, preservatives, emulsifiers, flavourings, colourings.

Students should be able to identify formulations given appropriate information — e.g. a label listing several ingredients with precise percentages indicates a formulation. You do **not** need to know the components of specific formulations.

> **Exam Tip:** A common 2-mark question: "Explain why a toothpaste is a formulation." Answer: it is a **mixture** (1) of components in **measured/specific quantities**, designed to give a **useful product** with particular properties (1).

4.8.1.3 Chromatography

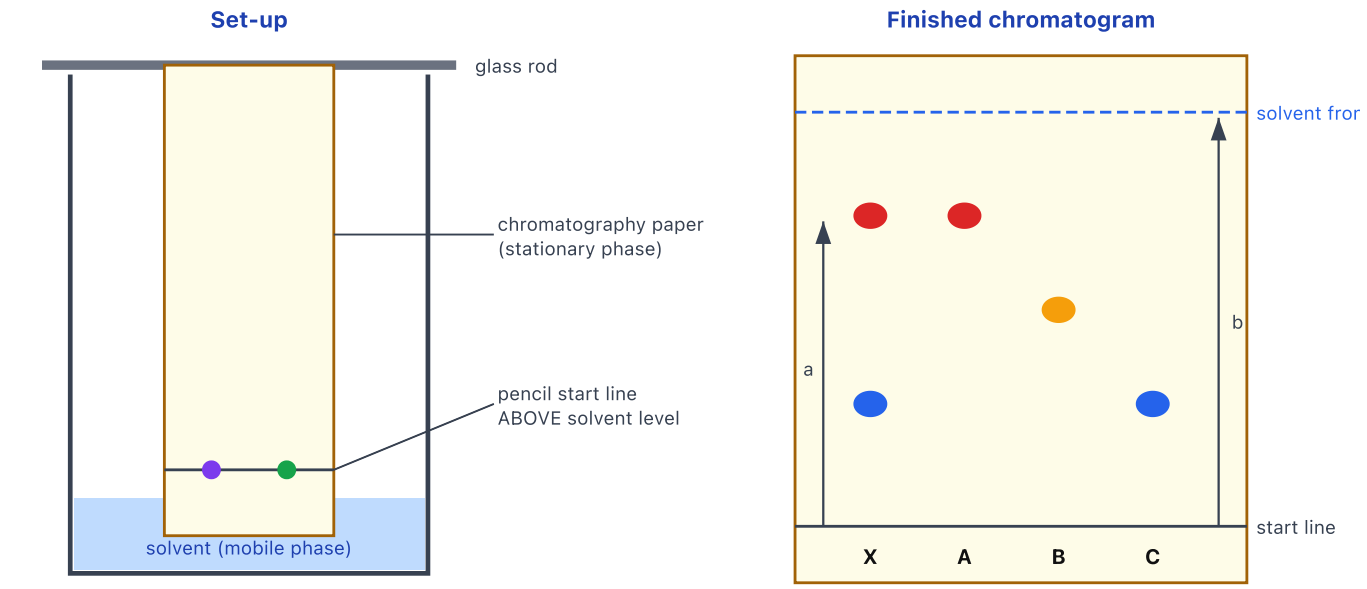
Chromatography can be used to **separate mixtures** and can give information to help **identify substances**. It is particularly useful for coloured substances such as inks, food colourings and dyes.

Chromatography involves a **stationary phase** and a **mobile phase**.

- In **paper chromatography**, the stationary phase is the **paper** (more precisely, water held in the paper fibres).
- The mobile phase is the **solvent** (e.g. water or ethanol) that moves up the paper.

Separation depends on the **distribution of substances between the phases**. A substance that is more soluble in the solvent and less attracted to the paper spends more time in the mobile phase, so it travels **further** up the paper.

Paper chromatography — set-up and chromatogram



Lid/watch glass on top reduces evaporation of solvent

R_f = a ÷ b. X contains A and C, but not B.

R_f Values

The ratio of the distance moved by a compound (centre of spot from origin) to the distance moved by the solvent can be expressed as its **R_f value**:

> R_f = distance moved by substance ÷ distance moved by solvent
>
> R_f has **no units** and is always **between 0 and 1**. Different compounds have different R_f values in different solvents, which can be used to help identify the compounds.

Key facts for interpreting chromatograms:

- A **pure** compound produces a **single spot** in all solvents.
- A mixture separates into **two or more spots** (though a single spot does not prove purity — two substances might have the same R_f in that solvent, so you should try a different solvent).
- If a spot in the unknown is at the **same height** (same R_f) as a spot from a known reference substance **in the same solvent**, they are likely to be the same compound.
- A substance that **stays on the start line** ($R_f = 0$) is **insoluble** in that solvent.

Worked Example 1 — Calculating an R_f Value

```
Question: On a chromatogram the solvent front moved 8.0 cm from
          the start line. A red spot moved 5.2 cm. Calculate its
          Rf value.

Given:   distance moved by substance = 5.2 cm
         distance moved by solvent   = 8.0 cm
Formula: Rf = distance moved by substance ÷ distance moved by solvent
Working: Rf = 5.2 ÷ 8.0
           = 0.65

Answer:  Rf = 0.65  (no units)
```

Worked Example 2 — Identifying Components

```
Question: A food colouring gives spots with Rf = 0.30 and 0.72.
          Reference dyes in the same solvent have Rf values:
            E102 = 0.72,  E110 = 0.45,  E124 = 0.30,  E133 = 0.85
          Which dyes are in the food colouring?

Answer:   The spots match E124 (0.30) and E102 (0.72).
```

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```
The colouring is a mixture of E124 and E102 – it does
not contain E110 or E133.
```

Required Practical — Chromatography

Aim: investigate how paper chromatography can be used to separate and tell the difference between coloured substances, and calculate R_f values.

1. Draw a **start line in pencil** about 2 cm from the bottom of the chromatography paper.
2. Use a capillary tube to put a **small, concentrated spot** of each ink/dye on the line, spaced apart. Label each in pencil.
3. Pour solvent into a beaker to a depth of about 1 cm — **below the start line**.
4. Hang the paper so that its bottom edge dips into the solvent but the spots are above the solvent. Put a lid on the beaker.
5. Allow the solvent to rise until it is near the top. Remove the paper and immediately **mark the solvent front** in pencil.
6. Let the paper dry. Measure distances from the start line to the **centre of each spot** and to the solvent front. Calculate R_f values.

Precaution	Reason
Start line drawn in **pencil**, not ink	Graphite is insoluble — it will not run up the paper. Ink would dissolve and separate, confusing the results.
Start line **above** the solvent level	Otherwise the samples would **dissolve into the solvent** in the beaker instead of travelling up the paper.
Lid on beaker	Keeps the atmosphere saturated with solvent vapour, so the solvent does not evaporate from the paper.
Small spots	Large spots spread out and overlap, making it hard to measure distances.
Paper must not touch the sides of the beaker	The solvent would move unevenly and the solvent front would not be straight.

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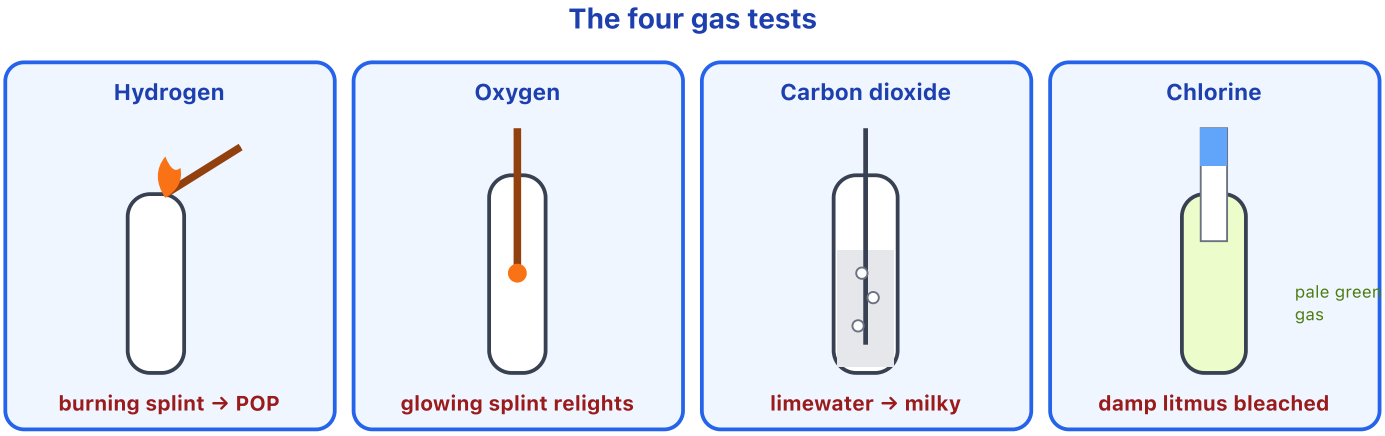
Exam Tip: The two most common chromatography exam questions are: "Why is the start line drawn in pencil?" and "Why must the start line be above the solvent?". Learn both answers word-for-word from the table above. Also remember: measure to the **centre** of the spot.

4.8.2 Identification of Common Gases

You must know the tests for four common gases. Each test uses a simple piece of equipment and gives a clear, distinctive result.

Gas	Test	Positive result	Chemistry behind it
Hydrogen (H_2)	Hold a **burning splint** at the open end of a test tube of the gas.	Hydrogen burns rapidly with a **squeaky pop**.	$2H_2 + O_2 \rightarrow 2H_2O$ (rapid combustion)
Oxygen (O_2)	Insert a **glowing splint** into a test tube of the gas.	The splint **relights**.	Oxygen supports combustion — the higher concentration restarts burning.
Carbon dioxide (CO_2)	Bubble the gas through (or shake it with) **limewater** (calcium hydroxide solution).	Limewater turns **milky (cloudy)**.	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$; insoluble calcium carbonate forms a white precipitate.
Chlorine (Cl_2)	Hold **damp litmus paper** in the gas.	Litmus paper is **bleached and turns white**. (Blue litmus may first turn red briefly.)	Chlorine dissolves in the water on the paper forming acidic, bleaching solutions.

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Exam Tip: The splints are easy to mix up: **burning** splint for **hydrogen** (pop), **glowing** splint for **oxygen** (relights). The litmus for chlorine must be **damp** — dry litmus will not work because the chlorine has to dissolve. And for CO_2, write "limewater turns **milky** / **cloudy**", never "limewater goes white".

4.8.3 Identification of Ions by Chemical and Spectroscopic Means (Chemistry only)

An ionic compound contains a **positive ion** (cation — usually a metal ion) and a **negative ion** (anion — a non-metal ion such as a halide, carbonate or sulfate). To identify the compound you need to carry out one test for the cation *and* one test for the anion.

4.8.3.1 Flame Tests

Flame tests can be used to identify some **metal ions (cations)**. Each metal ion gives a characteristic flame colour.

Metal ion	Flame colour
Lithium, Li^+	Crimson
Sodium, Na^+	Yellow
Potassium, K^+	Lilac

Calcium, Ca^{2+}	Orange-red
Copper, Cu^{2+}	Green (blue-green)

Method

1. Clean a **nichrome (or platinum) wire loop** by dipping it in concentrated hydrochloric acid and holding it in a **blue Bunsen flame** until no colour is seen.
2. Dip the loop into the solid sample (or solution).
3. Hold the loop in the **edge of a roaring blue flame** and observe the colour.
4. Clean the loop again before testing the next sample.

Limitations of Flame Tests

- If a sample contains a **mixture of ions**, some flame colours can be **masked**. For example, the bright yellow of sodium (even as a trace impurity) hides the faint lilac of potassium.
- Some colours are similar (lithium crimson and calcium orange-red) and judgement of colour is subjective.
- Flame tests are not quantitative — they do not show how much of an ion is present.

> **Exam Tip:** Learn the exact words AQA uses: lithium **crimson**, sodium **yellow**, potassium **lilac**, calcium **orange-red**, copper **green**. "Red" alone for lithium or "purple" for potassium may not be accepted. A mnemonic: "**Li**ttle **Cr**abs, **Na**sty **Y**ellow, **K**ind **L**ilac, **Ca**reful **O**range-red, **Cu**te **G**reen."

4.8.3.2 Metal Hydroxides

Sodium hydroxide solution can be used to identify some metal ions. Add a few drops of dilute NaOH(aq) to a solution of the unknown compound. Many metal hydroxides are **insoluble** and form a coloured **precipitate**.

Metal ion	Result with a few drops of NaOH(aq)	With excess NaOH(aq)

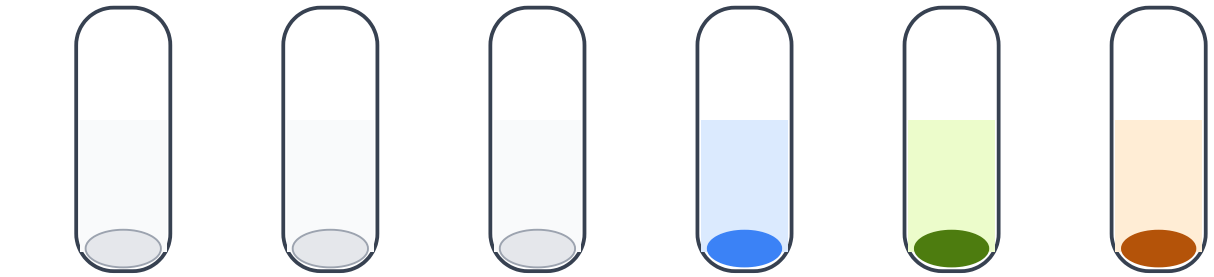
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Aluminium, Al^{3+}	**White** precipitate	Precipitate **dissolves** to give a colourless solution
Calcium, Ca^{2+}	**White** precipitate	Precipitate remains
Magnesium, Mg^{2+}	**White** precipitate	Precipitate remains
Copper(II), Cu^{2+}	**Blue** precipitate	Precipitate remains
Iron(II), Fe^{2+}	**Green** precipitate (turns orange-brown on standing in air)	Precipitate remains
Iron(III), Fe^{3+}	**Brown** (orange-brown / rusty) precipitate	Precipitate remains

Distinguishing the white precipitates:

- **Aluminium** is the only one whose precipitate **dissolves in excess** sodium hydroxide.
- **Calcium** and **magnesium** both stay as white precipitates. Use a **flame test** to tell them apart: calcium gives an **orange-red** flame; magnesium gives **no colour**.

Precipitates formed with sodium hydroxide solution



Equations for Hydroxide Precipitation

Students should be able to write balanced equations for the reactions that produce the insoluble hydroxides.

Word / symbol equations:

copper(II) sulfate + sodium hydroxide → copper(II) hydroxide + sodium sulfate
$CuSO_4(aq) + 2NaOH(aq) \rightarrow Cu(OH)_2(s) + Na_2SO_4(aq)$

iron(III) chloride + sodium hydroxide → iron(III) hydroxide + sodium chloride
$FeCl_3(aq) + 3NaOH(aq) \rightarrow Fe(OH)_3(s) + 3NaCl(aq)$

magnesium nitrate + sodium hydroxide → magnesium hydroxide + sodium nitrate
$Mg(NO_3)_2(aq) + 2NaOH(aq) \rightarrow Mg(OH)_2(s) + 2NaNO_3(aq)$

Ionic Equations for Hydroxide Precipitates (HT only)

Only the metal ion and the hydroxide ions take part; the other ions (e.g. Na^+, SO_4^{2-}) are **spectator ions** and are left out.

$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$	blue
$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$	green
$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$	brown
$Al^{3+}(aq) + 3OH^-(aq) \rightarrow Al(OH)_3(s)$	white
$Ca^{2+}(aq) + 2OH^-(aq) \rightarrow Ca(OH)_2(s)$	white
$Mg^{2+}(aq) + 2OH^-(aq) \rightarrow Mg(OH)_2(s)$	white

> **Exam Tip:** The number of OH^- ions in the ionic equation equals the **charge on the metal ion**. That is how you balance them: a 3+ ion needs $3OH^-$. Always include state symbols — the **(s)** on the hydroxide shows that it is a precipitate, which is the whole point of the test.

4.8.3.3 Carbonates

Carbonates react with **dilute acids** to form carbon dioxide gas. Carbon dioxide can be identified with **limewater**.

1. Add a few drops of **dilute hydrochloric acid** to the sample.
2. If it **fizzes**, bubble the gas through **limewater**.
3. Limewater turns **milky** → carbonate ions (CO_3^{2-}) are present.

```
sodium carbonate + hydrochloric acid → sodium chloride + water + carbon dioxide
Na₂CO₃(s) + 2HCl(aq) → 2NaCl(aq) + H₂O(l) + CO₂(g)

Ionic equation (HT):  CO₃²⁻ + 2H⁺ → H₂O + CO₂
```

4.8.3.4 Halides

Halide ions (Cl^-, Br^-, I^-) in solution produce precipitates with **silver nitrate solution** in the presence of **dilute nitric acid**.

1. Add a few drops of **dilute nitric acid** to the solution.
2. Then add a few drops of **silver nitrate solution**.

Halide ion	Precipitate	Colour	Ionic equation
Chloride, Cl^-	Silver chloride, AgCl	**White**	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
Bromide, Br^-	Silver bromide, AgBr	**Cream**	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$
Iodide, I^-	Silver iodide, AgI	**Yellow**	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$

```
Full equation example:
sodium chloride + silver nitrate → silver chloride + sodium nitrate
NaCl(aq) + AgNO₃(aq) → AgCl(s) + NaNO₃(aq)
```

Why add nitric acid first? The acid reacts with and removes any **carbonate ions**, which would otherwise also form a precipitate (silver carbonate) with silver nitrate and give a false positive. **Hydrochloric acid must not be used** because it would add chloride ions, giving a white precipitate whatever the sample.

4.8.3.5 Sulfates

Sulfate ions (SO_4^{2-}) in solution produce a **white precipitate** with **barium chloride solution** in the presence of **dilute hydrochloric acid**.

1. Add a few drops of **dilute hydrochloric acid**.

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2. Then add a few drops of **barium chloride solution**.
3. A **white precipitate** of barium sulfate → sulfate ions present.

```
sodium sulfate + barium chloride → barium sulfate + sodium chloride
Na₂SO₄(aq) + BaCl₂(aq) → BaSO₄(s) + 2NaCl(aq)

Ionic equation:  Ba²⁺(aq) + SO₄²⁻(aq) → BaSO₄(s)
```

Why add hydrochloric acid first? To remove **carbonate ions**, which would otherwise form a white precipitate of barium carbonate. **Sulfuric acid must not be used** because it contains sulfate ions and would always give a positive result.

Summary of Anion Tests

Anion	Reagent(s)	Positive result
Carbonate, CO_3^{2-}	Dilute acid, then gas into limewater	Fizzing; limewater turns milky
Chloride, Cl^-	Dilute HNO_3 then $AgNO_3(aq)$	White precipitate
Bromide, Br^-	Dilute HNO_3 then $AgNO_3(aq)$	Cream precipitate
Iodide, I^-	Dilute HNO_3 then $AgNO_3(aq)$	Yellow precipitate
Sulfate, SO_4^{2-}	Dilute HCl then $BaCl_2(aq)$	White precipitate

> **Exam Tip:** Pair the acid with the reagent by matching the anion: **silver nitrate** goes with **nitric** acid; **barium chloride** goes with **hydrochloric** acid. Using the "matching" acid ensures you never add the ion you are testing for. Six-mark questions often ask you to plan how to identify several unlabelled solutions — set out the tests in a logical sequence with expected results for each.

Required Practical — Identifying Ions

Aim: use chemical tests to identify the ions in unknown single ionic compounds — covering flame tests (Li^+, Na^+, K^+, Ca^{2+}, Cu^{2+}), hydroxide precipitates (Al^{3+}, Ca^{2+}, Mg^{2+}, Cu^{2+}, Fe^{2+}, Fe^{3+}), and tests for carbonate, halide and sulfate ions.

Worked Example 3 — Identifying an Unknown Compound

```
Question: A white solid gives a lilac flame. A solution of the solid
          gives a yellow precipitate when dilute nitric acid and
          silver nitrate solution are added. Identify the solid.

Cation:   lilac flame          → potassium ion, K⁺
Anion:    yellow ppt with Ag⁺  → iodide ion, I⁻

Answer:   potassium iodide, KI

Question: A green solution gives a green precipitate with NaOH(aq)
          and a white precipitate with HCl(aq) then BaCl₂(aq).

Cation:   green ppt with OH⁻   → iron(II), Fe²⁺
Anion:    white ppt with Ba²⁺  → sulfate, SO₄²⁻

Answer:   iron(II) sulfate, FeSO₄
```

Safety in Ion Tests

- Wear **eye protection** — sodium hydroxide is corrosive, even dilute solutions can damage eyes.
- Barium chloride is **toxic**; silver nitrate stains skin and is an irritant.
- Take care with the Bunsen flame and the hot wire loop; concentrated HCl used for cleaning is corrosive.

4.8.3.6 Instrumental Methods

Elements and compounds can be detected and identified using **instrumental methods** (machines). Examples include flame emission spectroscopy, mass spectrometry, infrared spectroscopy and gas chromatography.

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Compared with chemical tests, instrumental methods are:

- **Accurate** — reliably identify substances and give precise measurements of concentration.
- **Sensitive** — can detect and identify **very small amounts** of substance.
- **Rapid** — results are obtained very quickly and the process can be automated.

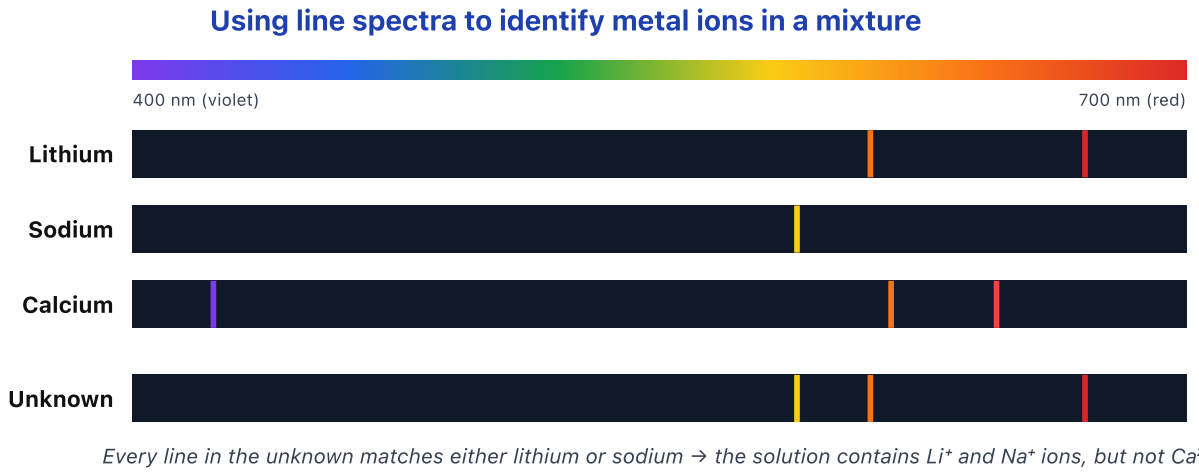
Instrumental methods	Chemical (wet) tests
Very accurate and sensitive; work on tiny samples	Need larger samples
Fast; can analyse many samples automatically	Slower; carried out by hand
Can identify mixtures and measure concentration	Usually only identify a single ion; colours can be masked
Expensive equipment; needs trained operators; results compared with a reference database	Cheap, simple equipment

4.8.3.7 Flame Emission Spectroscopy

Flame emission spectroscopy is an example of an instrumental method used to analyse **metal ions in solutions**.

1. The sample is put into a flame.
2. The metal ions absorb energy, and the light they give out is passed through a **spectroscope**.
3. The output is a **line spectrum** — a pattern of coloured lines at specific wavelengths.
4. The line spectrum can be analysed to **identify the metal ions** in the solution (each ion has its own unique spectrum, like a fingerprint) and to **measure their concentrations** (the brighter/more intense the lines, the higher the concentration).

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Every line in the unknown matches either lithium or sodium → the solution contains Li^+ and Na^+ ions, but not Ca^{2+}.

Exam Tip: The big advantage of flame emission spectroscopy over a simple flame test is that it can identify **several metal ions in a mixture** (no masking — each set of lines is separate) and can measure **concentration**. To interpret a spectrum question, check that **every** line of a reference metal appears in the unknown; if one line is missing, that metal is not present.

Worked Example 4 — Using a Calibration Line

```
Question: Standard solutions of sodium ions give these emission
          intensities (arbitrary units):
              Concentration (mg/dm³):   0    2    4    6    8
              Intensity:                0   15   30   45   60
          A sample of river water gives an intensity of 21.
          What is the concentration of sodium ions?

Method:  Intensity is directly proportional to concentration:
         15 units per 2 mg/dm³ → 7.5 units per 1 mg/dm³
         concentration = 21 ÷ 7.5 = 2.8 mg/dm³

Answer:  2.8 mg/dm³ (read from the calibration graph)
```

Summary Checklist for 4.8 Chemical Analysis

- Define a pure substance in chemistry and contrast it with the everyday meaning.
- Use melting and boiling point data to distinguish pure substances from mixtures.

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- Define and identify formulations (paints, medicines, fuels, alloys, fertilisers, foods, cleaning agents).
- Explain chromatography in terms of stationary and mobile phases; calculate and interpret R_f values.
- Describe the chromatography required practical and explain each precaution.
- Describe the tests for hydrogen, oxygen, carbon dioxide and chlorine and their positive results.
- *Chemistry only:* recall flame test colours for Li^+, Na^+, K^+, Ca^{2+} and Cu^{2+}, and explain why mixtures can mask colours.
- *Chemistry only:* recall the hydroxide precipitate colours for Al^{3+}, Ca^{2+}, Mg^{2+}, Cu^{2+}, Fe^{2+} and Fe^{3+}; write balanced (and, HT, ionic) equations.
- *Chemistry only:* describe the tests for carbonate, halide and sulfate ions, and explain the choice of acid in each.
- *Chemistry only:* state the advantages of instrumental methods (accurate, sensitive, rapid).
- *Chemistry only:* describe flame emission spectroscopy and interpret line spectra and calibration data.

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